

Appendix

Part I. Data collection

Dataset I: ABR testing was conducted in a sound-treated chamber using the Eclipse EP 25 Evoked Potential System (Interacoustics Inc., Denmark) as part of clinical routine audiological assessment. The stimuli were 100 μ s clicks presented in alternating polarity at a rate of 20.1/s, 31.5/s or 44.4/s. Four surface electrodes were placed at Cz, middle forehead (ground), and bilateral mastoids (references). Responses were filtered at 0.033–3 kHz, sampled at 30 kHz, with the artifact rejection level set at 40 μ V. Each waveform comprised 360 points spanning 0–12 ms post-stimulus, averaged from $\sim$ 1,250 artifact-free epochs (SD: $\sim$ 600).

Dataset II: ABRs were elicited by 100 μ s clicks at 11.1/s, 19.3/s, or 21.1/s, recorded using the SmartEP system (Intelligent Hearing Systems Inc., USA) with electrode placement identical to that in Dataset I. Responses were sampled at 40 kHz and filtered at 0.1–3 kHz, with a 40 μ V artifact rejection threshold. Recordings (0–12.8 ms) were clipped to 0–12 ms and down-sampled to 30 kHz to match Dataset I.

Mouse Dataset: Stimuli included 0.01-ms clicks and 5-ms tone pips (1-ms rise/fall) at 6, 12, 18, 24, and 30 kHz, presented at 42.6/s in 5-dB steps over specified SPL ranges (6 kHz: 20 to 85 dB; 12 kHz: 0 to 70 dB; 18 kHz: 0 to 70 dB; 24 kHz: 10 to 70 dB; 30 kHz: 20 to 85 dB; clicks: 0 to 85 dB), with upper limits extended to 95 dB SPL for suspected hearing loss. Each response was averaged over 256 stimuli.

Part II. Supplemental Tables

Table A1. Example discretization rules for threshold labels and stimulus levels.

The same mapping strategy is applied to both, as ABR thresholds are defined based on stimulus intensity.

Threshold category	Stimulus level (dB nHL)	Encoded value
/	Zero-padding	11
NT	/	10
100	100	9
90	90	8
80	80	7
70	70	6
60	60	5
50	50	4
40	40	3
30	30	2
20	20	1
10	10	0
/	0	-1

Note: NT = No threshold, defined as the absence of a detectable response even at the maximum intensity (100 dB nHL).

Table A2. Demographic Characteristics For Both Human Datasets.

Characteristic		Dataset I (<i>n</i> = 8350)	Dataset II (<i>n</i> =136)
Age	<6 months	1118	14
	6 months-18 years	3961	85
	18-60 years	2409	30
	>60 years	862	7
Gender	Males	4654	90
	Females	3696	46
Grades of hearing loss	Normal hearing	499	9
	Mild hearing loss	583	5
	Moderate hearing loss	724	20
	Moderately severe hearing loss	846	12
	Severe hearing loss	807	10
	Profound hearing loss	635	9
	Total hearing loss	1017	11
	Unknown	3239	60

Note: The grades of hearing loss was defined according to the average hearing thresholds at 0.5, 1, 2, and 4 kHz in the better ear, categorized as follows: normal hearing: < 20 dB ; Mild hearing loss: $20 \text{ dB} \leq \text{threshold} < 35 \text{ dB}$; Moderate hearing loss: $35 \text{ dB} \leq \text{threshold} < 50 \text{ dB}$; Moderately severe hearing loss: $50 \text{ dB} \leq \text{threshold} < 65 \text{ dB}$; Severe hearing loss: $65 \text{ dB} \leq \text{threshold} < 80 \text{ dB}$; Profound hearing loss: $80 \text{ dB} \leq \text{threshold} < 95 \text{ dB}$; Complete or total hearing loss: $95 \text{ dB} \leq \text{threshold}$; Unknown: not tested.

Table A3. Age Distributions of The Subgroup Without Available Hearing Information For both Human Datasets

Age, <i>n</i> (%)	Dataset I	Dataset II
< 6 months	953 (29.4%)	14 (23.3%)
6 months - 3 years	1566 (48.3%)	26 (43.3%)
3 - 6 years	444 (13.7%)	17 (28.3%)
6 - 12 years	53 (1.6%)	1 (1.7%)
12-18 years	13 (0.4%)	0 (0)
> 18 years	210 (6.5%)	2 (3.3%)
Total	3239	60

Table A4. The optimal hyperparameters used in this work.

		Series Transformer			Temporal Transformer					
		Series	Series	Series	Stage 1		Stage 2		Stage 3	
		Encode r 1	Encoder 2	Encode r 3	Temporal slice	Encoder	Temporal slice	Encoder	Temporal slice	Encoder
Dataset I		$d_k = 128, H_s = 4, L_s = 6$			$k^{(1)} = 16,$ $f^{(1)} = 64,$ $s^{(1)} = 4$	$L_1 = 4,$ $H_T^{(1)} = 4,$ $R^{(1)} = 2$	$k^{(2)} = 8,$ $f^{(2)} = 64,$ $s^{(2)} = 2$	$L_2 = 4,$ $H_T^{(2)} = 4,$ $R^{(2)} = 1$	$k^{(3)} = 4$ $f^{(3)} = 64,$ $s^{(3)} = 1$	$L_3 = 4,$ $H_T^{(3)} = 4,$ $R^{(3)} = 1$
Mouse Dataset	Click	$d_k = 64, H_s = 4, L_s = 6$			$k^{(1)} = 16,$ $f^{(1)} = 64,$ $s^{(1)} = 4$	$L_1 = 4,$ $H_T^{(1)} = 8,$ $R^{(1)} = 2$	$k^{(2)} = 8,$ $f^{(2)} = 64,$ $s^{(2)} = 2$	$L_2 = 4,$ $H_T^{(2)} = 8,$ $R^{(2)} = 1$	$k^{(3)} = 4$ $f^{(3)} = 64,$ $s^{(3)} = 1$	$L_3 = 4,$ $H_T^{(3)} = 8,$ $R^{(3)} = 1$
	6KHz	$d_k = 64, H_s = 4, L_s = 4$			$k^{(1)} = 8,$ $f^{(1)} = 64,$ $s^{(1)} = 4$	$L_1 = 4,$ $H_T^{(1)} = 8,$ $R^{(1)} = 2$	$k^{(2)} = 4,$ $f^{(2)} = 64,$ $s^{(2)} = 2$	$L_2 = 4,$ $H_T^{(2)} = 8,$ $R^{(2)} = 1$	$k^{(3)} = 2,$ $f^{(3)} = 64,$ $s^{(3)} = 1$	$L_3 = 4,$ $H_T^{(3)} = 8,$ $R^{(3)} = 1$
	12KHz	$d_k = 64, H_s = 8, L_s = 6$			$k^{(1)} = 16,$ $f^{(1)} = 128,$ $s^{(1)} = 4$	$L_1 = 2,$ $H_T^{(1)} = 4,$ $R^{(1)} = 2$	$k^{(2)} = 4,$ $f^{(2)} = 128,$ $s^{(2)} = 2$	$L_2 = 2,$ $H_T^{(2)} = 4,$ $R^{(2)} = 1$	$k^{(3)} = 2,$ $f^{(3)} = 128,$ $s^{(3)} = 1$	$L_3 = 2,$ $H_T^{(3)} = 4,$ $R^{(3)} = 1$
	18KHz	$d_k = 64, H_s = 4, L_s = 6$			$k^{(1)} = 16,$ $f^{(1)} = 128,$ $s^{(1)} = 4$	$L_1 = 4,$ $H_T^{(1)} = 4,$ $R^{(1)} = 2$	$k^{(2)} = 4,$ $f^{(2)} = 128,$ $s^{(2)} = 2$	$L_2 = 4,$ $H_T^{(2)} = 4,$ $R^{(2)} = 1$	$k^{(3)} = 2,$ $f^{(3)} = 128,$ $s^{(3)} = 1$	$L_3 = 4,$ $H_T^{(3)} = 4,$ $R^{(3)} = 1$
	24KHz	$d_k = 64, H_s = 4, L_s = 6$			$k^{(1)} = 4,$ $f^{(1)} = 128,$ $s^{(1)} = 4$	$L_1 = 4,$ $H_T^{(1)} = 4,$ $R^{(1)} = 2$	$k^{(2)} = 2,$ $f^{(2)} = 128,$ $s^{(2)} = 2$	$L_2 = 4,$ $H_T^{(2)} = 4,$ $R^{(2)} = 1$	$k^{(3)} = 1,$ $f^{(3)} = 128,$ $s^{(3)} = 1$	$L_3 = 4,$ $H_T^{(3)} = 4,$ $R^{(3)} = 1$
	30KHz	$d_k = 128, H_s = 4, L_s = 6$			$k^{(1)} = 16,$ $f^{(1)} = 128,$ $s^{(1)} = 4$	$L_1 = 2,$ $H_T^{(1)} = 8,$ $R^{(1)} = 2$	$k^{(2)} = 8,$ $f^{(2)} = 128,$ $s^{(2)} = 2$	$L_2 = 2,$ $H_T^{(2)} = 8,$ $R^{(2)} = 1$	$k^{(3)} = 4,$ $f^{(3)} = 128,$ $s^{(3)} = 1$	$L_3 = 2,$ $H_T^{(3)} = 8,$ $R^{(3)} = 1$
	Tone pips combined	$d_k = 128, H_s = 4, L_s = 6$			$k^{(1)} = 8,$ $f^{(1)} = 128,$ $s^{(1)} = 4$	$L_1 = 4,$ $H_T^{(1)} = 8,$ $R^{(1)} = 2$	$k^{(2)} = 4,$ $f^{(2)} = 128,$ $s^{(2)} = 2$	$L_2 = 4,$ $H_T^{(2)} = 8,$ $R^{(2)} = 1$	$k^{(3)} = 2,$ $f^{(3)} = 128,$ $s^{(3)} = 1$	$L_3 = 4,$ $H_T^{(3)} = 8,$ $R^{(3)} = 1$

d_k : the dimension of Q and K in the Series Encoder.

H_s : the number of attention heads of Dynamic Time Warping Attention

$k^{(l)}$: the kernel size of stage l .

$f^{(l)}$: the number of filters of stage l .

$s^{(l)}$: the stride size of stage l .

$R^{(l)}$: the temporal reduction rate of stage l .

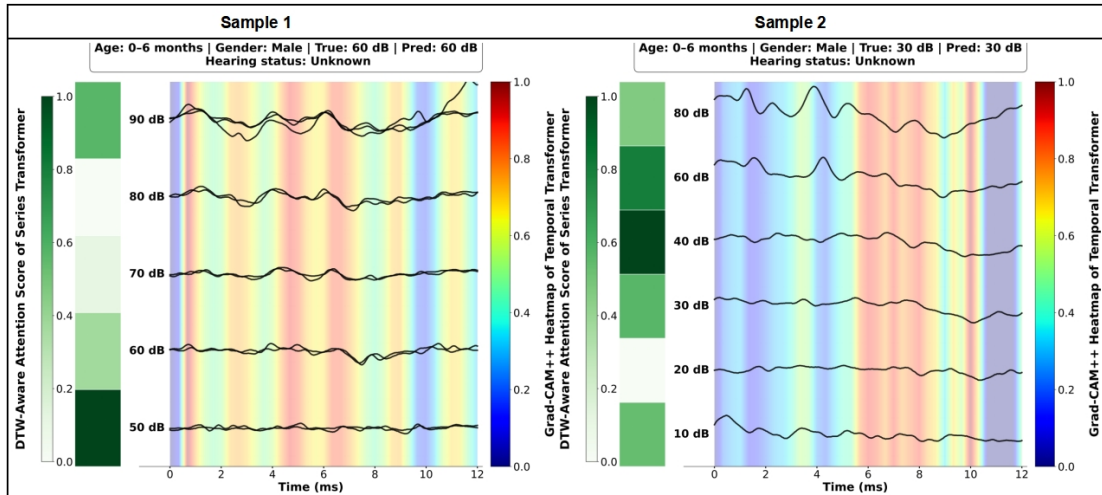
$H_T^{(l)}$: the number of attention heads of Temporal Reduction Attention of stage l .

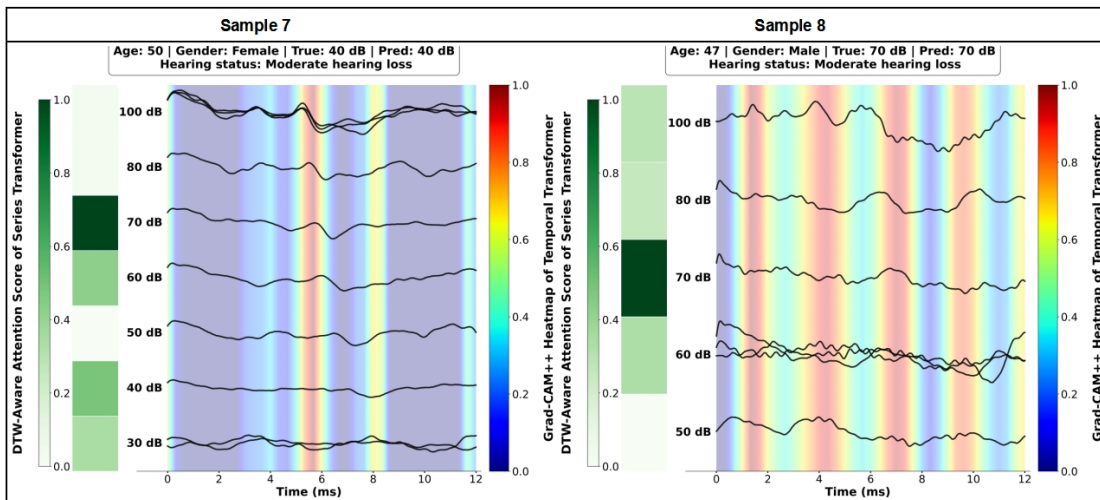
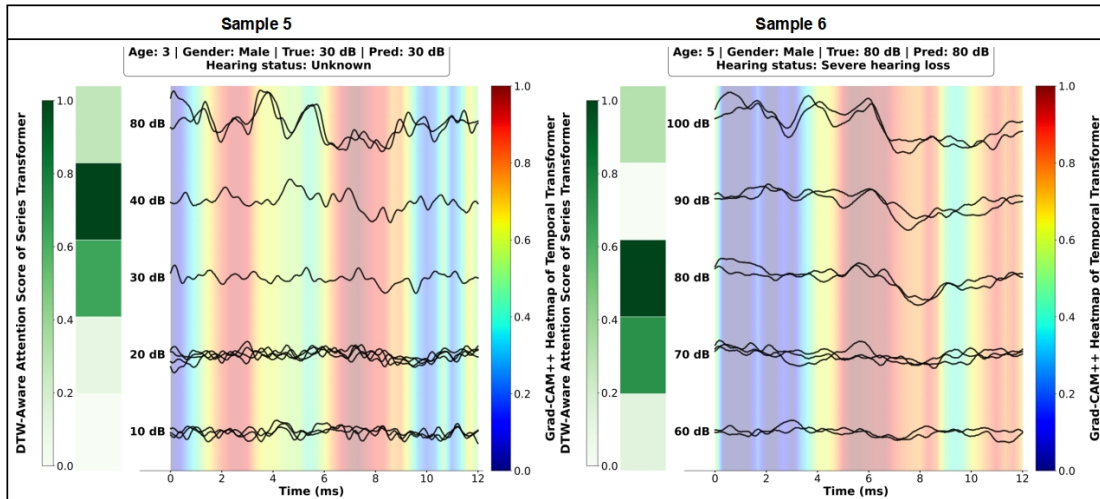
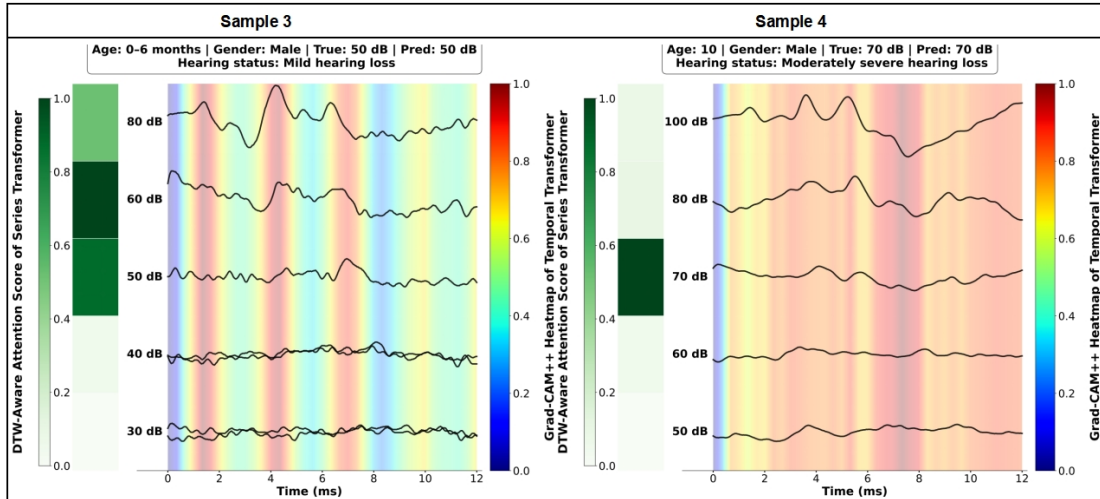
L_l : the number of Temporal encoder layers of stage l .

L_s : the number of Series encoder layers.

Part III. Interpretability Analysis of DTWA-STformer

To better understand the decision-making process of DTWA-STformer, we performed visualization analyses on ABR stacks. First, to illustrate the distribution of contributions from waveforms at different stimulus levels to the final threshold prediction, we visualized the attention scores from the last Series Encoder. Specifically, the attention score matrix from the DTW-attn module was extracted, averaged across heads, and further row-averaged to obtain a one-dimensional importance vector across waveform levels. This vector reflected the relative importance of each valid ABR waveform within the stack to the final threshold prediction. Second, to examine whether the Temporal Transformer focused on clinically meaningful temporal regions, we applied Gradient-weighted Class Activation Mapping++ (Grad-CAM++) to the final Temporal Encoder layer. The generated activation maps were upsampled to the original temporal resolution of the ABR signals and overlaid on the raw waveforms. The visualizations for ten representative cases are presented in Figure A1. Overall, the Series Transformer generally assigns greater importance to waveforms at threshold or near-threshold levels. The Temporal Transformer mainly focuses on clinically relevant peak regions, especially around wave V, and in some cases also earlier peak regions (e.g., waves III and I) that may assist threshold judgment. These patterns are broadly consistent with experts' ABR thresholding process.





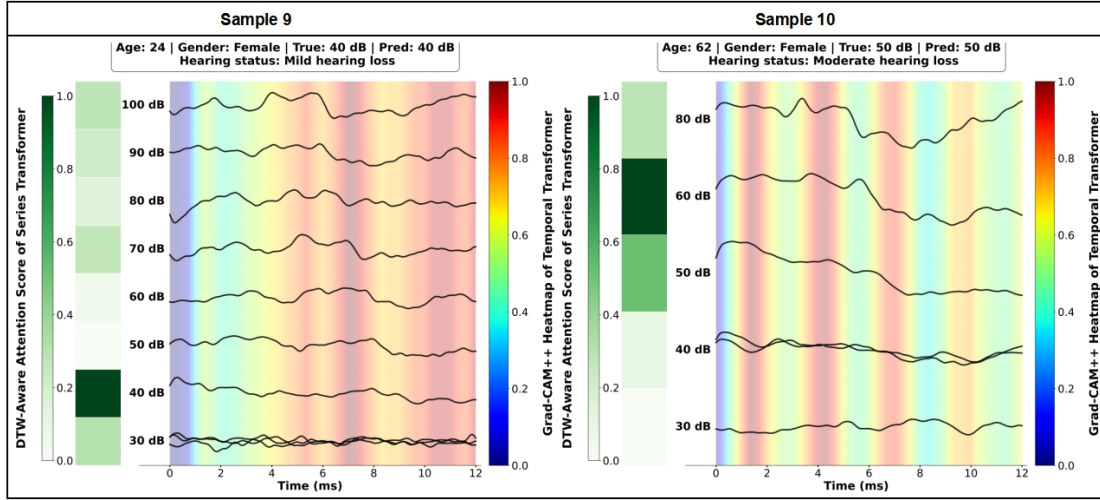


Figure A1. Interpretability visualizations of DTWA-STformer on ten representative human ABR stacks. For each stack, the green bar on the left shows the waveform-level importance scores derived from the DTW-aware Series Transformer, reflecting the relative contribution of ABR waveforms at different stimulus intensities to the final threshold prediction, with darker colors indicating greater importance. The overlaid heatmap on the right shows the Grad-CAM++ visualization of the Temporal Transformer, highlighting the temporal regions that contribute most strongly to the model’s decision, with warmer colors indicating higher relevance.